EECE5576: Wireless Communication Systems: Homework Assignment 2

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Submission Instructions:
Submit your assignment in electronic format (.pdf) on Canvas, named 'LastName_HW2.pdf'.

Use of AI tools in this assignment. This assignment has two regimes, and they are graded differently.

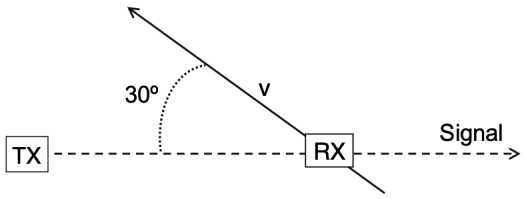
- **Problem 1 (a), (b) and Problem 2: AI tools are permitted** as a resource, on the same terms as any other resource. What is assessed here is whether you can *verify* and *justify* what you submit. Every submission must therefore include the **Verification Appendix** described at the end of this assignment, whether or not you used an AI tool. All explanations must be written in your own words.
- **Problem 1 (c): AI tools are not permitted.** The skill being assessed in (c) *is* the audit. Using an AI tool to perform the audit defeats the purpose of the assessment.

Submitting output you cannot explain, including fabricated references or results you cannot justify, is an academic integrity violation and will be handled under the Academic Integrity policy in the syllabus. The instructor may ask any student for a short (3–5 minute) oral walkthrough of any part of this submission.

NOTE: For all the problems, show all your calculations. If you use results and expressions not discussed during the lectures, indicate the source you have used as a reference.

Problem 1: A transmitter sends a signal with frequency 1800 MHz and bandwidth 400 MHz. The receiver is moving with respect to the transmitter at 60 km/h.

a) (15%) The system is observed at a given time. Consider LOS and consider only the direct signal received. The receiver is moving in a direction as shown in the figure:



During the observation time, the direction from which the signal arrives to the receiver does not change. Compute the Doppler shift and the received signal frequency in the receiver at the observation time.

b) (15%) Briefly explain the concept of coherence time. Consider a general scenario and a long observation time, calculate the coherence time of the channel perceived by the received signal.

c) (25%) **Audit of a proposed solution: average fading duration.** *No AI tools may be used in this part.*

Now consider a rich multipath (NLOS) scenario for the same receiver, so that the envelope follows a Rayleigh distribution with a Jakes Doppler spectrum. A *fade* is defined as the envelope falling below a threshold r_s that is **10 dB below the rms envelope level** r_{rms}.

A generative AI tool was asked to compute the average fading duration $D_F(r_s)$ for this receiver and to interpret the result. Its answer is reproduced below. The steps are labeled S1–S7 for reference. The final number looks plausible. **The solution is not correct.**

Proposed solution (AI-generated).

S1. The maximum Doppler frequency is $\nu_{max} = \dfrac{v f_c}{c} = \dfrac{(60)(1.8 \times 10^9)}{3 \times 10^8} = 360$ Hz.

S2. The signal bandwidth (400 MHz) does not enter the calculation, since D_F depends only on ν_{max} and on the normalized threshold ρ.

S3. The rms envelope level is $r_{rms} = \sqrt{2}\,\sigma$, where σ^2 is the variance of each of the in-phase and quadrature Gaussian components.

S4. The threshold is 10 dB below the rms level, so $\rho = r_s / r_{rms} = 10^{-10/10} = 0.1$.

S5. Using the Jakes-spectrum result for Rayleigh fading,

$$D_F(r_s) = \frac{e^{\rho^2} - 1}{\sqrt{2\pi}\,\nu_{max}\,\rho} = \frac{e^{0.01} - 1}{\sqrt{2\pi}\,(360)(0.1)} = \frac{0.01005}{90.24} \approx 1.11 \times 10^{-4} \text{ s} = 0.11 \text{ ms}.$$

S6. Since an average fade lasts only 0.11 ms, the fraction of time the signal spends below the threshold is approximately 1.1×10^{-4}, i.e. about 0.011%. A 10 dB fading margin is therefore far larger than this receiver needs.

S7. Because $D_F \propto 1/\nu_{max}$ for a fixed ρ, doubling the receiver speed halves the average fade duration at the same threshold.

In your report:

i. Identify the errors. For *each* error, state: (1) the step where it occurs; (2) what is wrong, in one or two sentences; (3) what the correct version is; (4) **the evidence that reveals it** (a unit check, a hand calculation, a physical argument, an inconsistency with your answers to (a) and (b), or an inconsistency with the lecture notes). Item (4) carries the most weight.

ii. The number of errors is not given to you. Not every step is wrong: if you believe a step is correct, say so and justify it. Flagging a correct step as incorrect will lose points.

iii. Give the corrected value of $D_F(r_s)$. Then compute the level crossing rate $N(r_s)$ at the same threshold and use it, together with $D_F(r_s)$, to obtain the fraction of time the signal is below r_s. Compare this fraction with the Rayleigh CDF evaluated at r_s, and use the comparison to confirm or reject S6.

iv. The incorrect value in S5 is of a plausible order of magnitude. Explain why a plausible-looking number is not, by itself, evidence that a solution is correct.

Problem 2: A receiver is receiving a signal from a transmitter. The carrier frequency transmitted is 2 GHz with bandwidth 600 kHz. The fading suffered by this signal can be considered flat fading for the whole transmission. The receiver is moving at a velocity of 60 km/hour.

a) (15%) Explain briefly the concept of flat fading and when a channel can be considered a flat fading channel. Is the signal in this flat fading channel distorted due to delay dispersion?

b) (20%) Find the limitations of delay spread that this signal is suffering in this scenario. Justify every step to reach your solution. If you use results not covered in the lectures, indicate the source you have used as a reference.

Verification Appendix (required, part of the 10% below). At the end of your report, include a short appendix containing:

1. A statement of whether you used an AI tool for Problem 1 (a), (b) or Problem 2, which tool, and for what. If you did not use one, say so.
2. The prompts you used, verbatim, if any.
3. One thing that was wrong, or that you had to change, in the first version of your solution, and how you detected it.

NOTE: (10%) Write a clear and organized report, include the Verification Appendix, and be organized in your submission.